

WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main(){
    char command[100];
    pid_t pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    } else {
        wait(NULL);
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }
    return 0;
}
```

Output:

```
Enter a Linux command: ls

Child Process
Child PID : 437
OSSP code.c files.c gm.c hello.c os program.c programming.c source_code
code files gm hello operating os.c programming programming.c.save source_code.c

Parent Process
Parent PID : 436
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

1. Uname: Displays operating system and kernel information.

```
rishith@RISHITH:~$ ./$ uname
Linux
rishith@RISHITH:~$ ./$ |
```

2. lscpu: Displays CPU information.

```
rishith@RISHITH:~$ ./$ lscpu
```

```
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: GenuineIntel
Model name: 12th Gen Intel(R) Core(TM) i5-12450H
CPU family: 6
Model: 154
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 3
BogoMIPS: 4991.99
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtr
r pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcmt tsc_deadline_timer aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch ssbd i
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrm md_clear serialize ibt flu
sh_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
```

```

sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fstrm md_clear serialize ibt flu
sh_lld arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 288 KiB (6 instances)
  L1i: 192 KiB (6 instances)
  L2: 7.5 MiB (6 instances)
  L3: 12 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
rishith@RISHITH:~$ .$.

```

3. lsblk: Displays storage devices.

```

rishith@RISHITH:~$ .$. lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 356.9M  1 disk
sdb         8:16    0 159.4M  1 disk
sdc         8:32    0     2G   0 disk [SWAP]
sdd         8:48    0     1T   0 disk /snap
                               /mnt/wslg/distro
                               /
rishith@RISHITH:~$ .$.

```

4. ps: Displays currently running processes.

```

rishith@RISHITH:~$ . $ ps
      PID TTY          TIME CMD
      382 pts/0        00:00:00 bash
      626 pts/0        00:00:00 ps
rishith@RISHITH:~$ . $

```

5. top: Displays real-time system performance.

```

rishith@RISHITH:~$ . $ top
top - 09:40:06 up 4 min,  1 user,  load average: 0.06, 0.08, 0.03
Tasks:  29 total,   1 running, 28 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni, 99.9 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7794.6 total,  6468.8 free,   542.7 used,   934.9 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,    0.0 used,  7252.0 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+
    1 root        20   0   21628 13156 9796 S   0.0   0.2   0:00.41
    2 root        20   0    3180  2212 2072 S   0.0   0.0   0:00.01
    6 root        20   0    3196  2132 2008 S   0.0   0.0   0:00.00
   52 root        19  -1  42164 16232 15156 S   0.0   0.2   0:00.09
   99 root        20   0   25336  6736 5124 S   0.0   0.1   0:00.12
  111 root        20   0  152944  1768 1512 S   0.0   0.0   0:00.00
  114 root        20   0  153076  1656 1396 S   0.0   0.0   0:00.00
  116 root        20   0  153076  1828 1512 S   0.0   0.0   0:00.00
  125 root        20   0  153076  1704 1444 S   0.0   0.0   0:00.00
  130 root        20   0  153076  1680 1424 S   0.0   0.0   0:00.00
  135 root        20   0  526764 12536 1536 S   0.0   0.2   0:01.90
  160 systemd+    20   0   21460 13192 10932 S   0.0   0.2   0:00.06
  162 systemd+    20   0   91028  7944  6976 S   0.0   0.1   0:00.04
  217 root        20   0    4244  2684  2432 S   0.0   0.0   0:00.00
  218 message+    20   0    9632  5304  4600 S   0.0   0.1   0:00.03
  227 root        20   0 2147016 42272 24680 S   0.0   0.5   0:00.37
  231 root        20   0   17964  8852  7812 S   0.0   0.1   0:00.04
  233 root        20   0 1756104 12996 10888 S   0.0   0.2   0:00.07
  247 syslog      20   0   222516  5976  4572 S   0.0   0.1   0:00.04
  256 root        20   0    3124  1976  1836 S   0.0   0.0   0:00.00
  270 root        20   0  107020 23188 13728 S   0.0   0.3   0:00.08
  380 root        20   0    3196  1120   980 S   0.0   0.0   0:00.00
  381 root        20   0    3196  1252  1108 S   0.0   0.0   0:00.01
  382 shivam      20   0    7568  6144  4372 S   0.0   0.1   0:00.03
  383 root        20   0    6704  4648  3864 S   0.0   0.1   0:00.00
  427 shivam      20   0   20376 11652  9492 S   0.0   0.1   0:00.06
  428 shivam      20   0   21152  3576  1852 S   0.0   0.0   0:00.00
  454 shivam      20   0    6080  5360  3692 S   0.0   0.1   0:00.01
  629 shivam      20   0    9300  5760  3572 R   0.0   0.1   0:00.01

```

